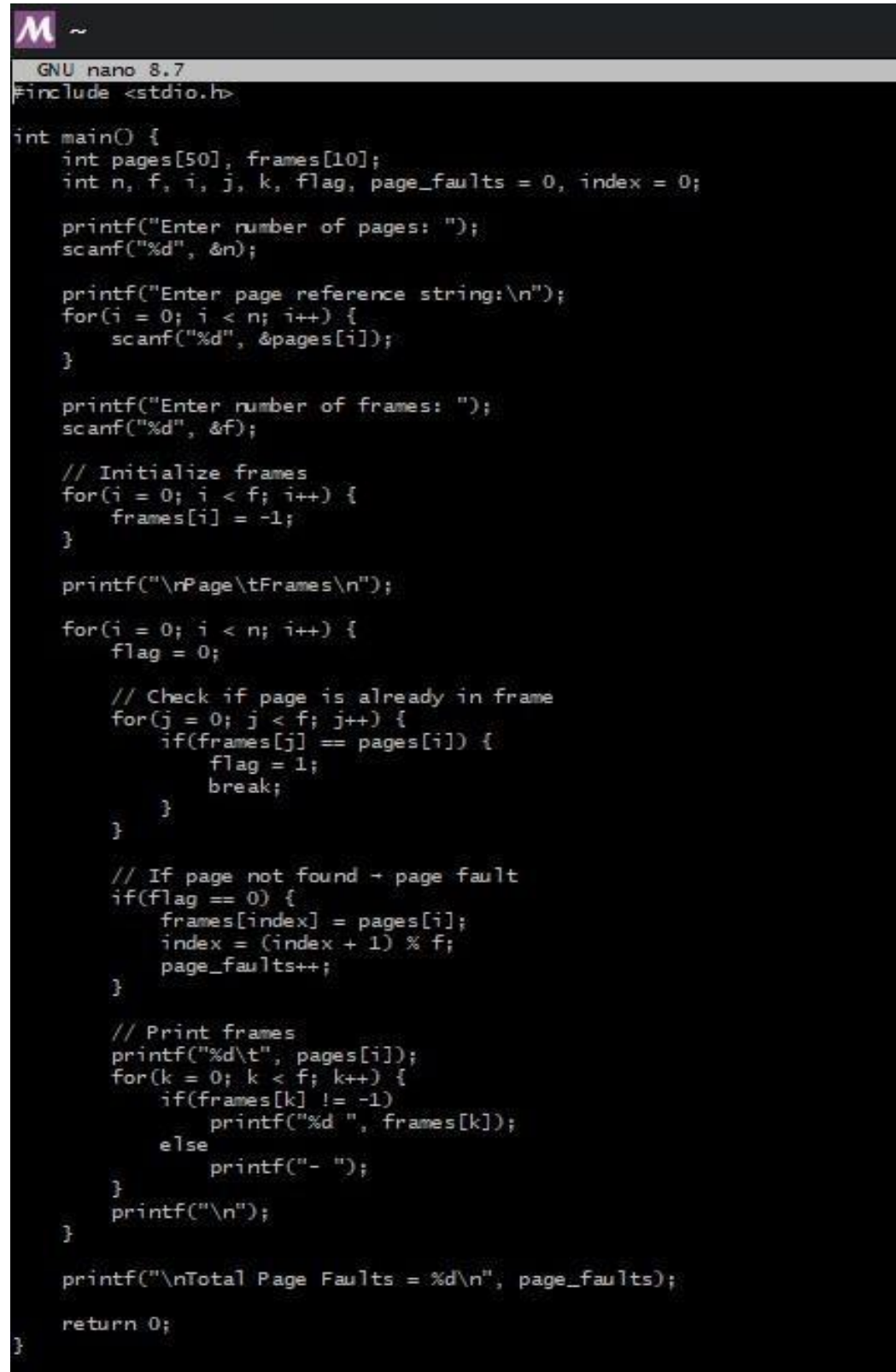


Practical 11 (Post-lab):

Aim: Implementing a Virtual Memory System with Demand Paging where pages are loaded into memory only when needed, optimizing memory usage and minimizing page faults.

Code:



```
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, f, i, j, k, flag, page_faults = 0, index = 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    scanf("%d", &f);

    // Initialize frames
    for(i = 0; i < f; i++) {
        frames[i] = -1;
    }

    printf("\rPage\tFrames\n");

    for(i = 0; i < n; i++) {
        flag = 0;

        // Check if page is already in frame
        for(j = 0; j < f; j++) {
            if(frames[j] == pages[i]) {
                flag = 1;
                break;
            }
        }

        // If page not found -> page fault
        if(flag == 0) {
            frames[index] = pages[i];
            index = (index + 1) % f;
            page_faults++;
        }

        // Print frames
        printf("%d\t", pages[i]);
        for(k = 0; k < f; k++) {
            if(frames[k] != -1)
                printf("%d ", frames[k]);
            else
                printf("- ");
        }
        printf("\n");
    }

    printf("\nTotal Page Faults = %d\n", page_faults);

    return 0;
}
```

Output:

```
M ~  
Admin@DESKTOP-PNDN9B0 MSYS ~  
$ nano practical11_demand.c  
  
Admin@DESKTOP-PNDN9B0 MSYS ~  
$ gcc practical11_demand.c -o demand  
./demand  
Page      Frames  
1          1 - -  
2          1 2 -  
3          1 2 3  
4          4 2 3  
1          4 1 3  
2          4 1 2  
5          5 1 2  
1          5 1 2  
2          5 1 2  
3          5 3 2  
4          5 3 4  
5          5 3 4  
  
Total Page Hits = 3  
Total Page Faults = 9  
  
Admin@DESKTOP-PNDN9B0 MSYS ~  
$ |
```